

Enriching stock market data using Open AI API

Market Summary > Nasdaq-100

15,101.71

+4,239.07 (39.02%) ↑ year to date

Aug 9, 5:15 PM EDT • Disclaimer

1D | 5D | 1M | 6M | YTD | 1Y | 5Y | Max



Open	15,286.99	Low	15,077.91	52-wk high	15,932.05
High	15,288.76	Prev close	15,273.05	52-wk low	10,440.64

The Nasdaq-100 is a stock market index made up of 101 equity securities issued by 100 of the largest non-financial companies listed on the Nasdaq stock exchange. It helps investors compare stock prices with previous prices to determine market performance.

In this project you are provided with two CSV files containing Nasdaq-100 stock information:

- **nasdaq100_CA.csv**: contains information about companies in the index such as symbol, name, etc. For this analysis, only companies headquartered in California have been selected.
- **nasdaq100_price_change.csv**: contains price changes per stock across periods including (but not limited to) one day, five days, one month, six months, one year, etc.

As an AI developer, you will leverage the OpenAI API to classify companies into sectors and produce a summary of sector and company performance for this year, for the companies in the index that are headquartered in California.

CSV with Nasdaq-100 stock data

In this project, you have available two CSV files `nasdaq100_CA.csv` and `nasdaq100_price_change.csv`.

nasdaq100_CA.csv

```
symbol,name,headQuarter,dateFirstAdded,cik,founded
AAPL,Apple Inc.,"Cupertino, CA",,0000320193,1976-04-01
ABNB,Airbnb,"San Francisco, CA",,0001559720,2008-08-01
ADBE,Adobe Inc.,"San Jose, CA",,0000796343,1982-12-01
...
```

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nasdaq100_price_change.csv

```
symbol,1D,5D,1M,3M,6M,ytd,1Y,3Y,5Y,10Y,max
AAPL,-1.7254,-8.30086,-6.20411,3.042,15.64824,42.99992,8.47941,60.96299,245.42031,976.99441
,139245.53954
ABNB,2.1617,-2.21919,9.88336,19.43286,19.64241,68.66902,23.64013,-1.04347,-1.04347,-1.04347
,-1.04347
ADBE,0.5409,-1.77817,9.16191,52.0465,38.01522,57.22723,21.96206,17.83037,109.05718,1024.692
14,251030.66399
ADI,0.9291,-4.03352,2.58486,3.65887,5.01602,17.02062,8.09735,63.42847,92.81874,286.77518,26
012.63736
...
```

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```

# Start your code here!
import os
import pandas as pd
from openai import OpenAI

# Instantiate an API client
client = OpenAI()

# Read in the two CSV files
nasdaq100_ca = pd.read_csv('nasdaq100_CA.csv')
nasdaq100_price_change = pd.read_csv('nasdaq100_price_change.csv')

# Merge the ytd column into nasdaq100_ca
nasdaq100_ca = nasdaq100_ca.merge(nasdaq100_price_change[['symbol', 'ytd']], on='symbol')

# Preview the result
print(nasdaq100_ca.head())

```

	symbol	name	headQuarter	...	cik	founded	ytd
0	AAPL	Apple Inc.	Cupertino, CA	...	320193	1976-04-01	42.99992
1	ABNB	Airbnb	San Francisco, CA	...	1559720	2008-08-01	68.66902
2	ADBE	Adobe Inc.	San Jose, CA	...	796343	1982-12-01	57.22723
3	ADSK	Autodesk	San Rafael, CA	...	769397	1982-01-30	10.02701
4	AMAT	Applied Materials	Santa Clara, CA	...	6951	1967-11-10	55.46366

```
[5 rows x 7 columns]
```

```

# Create a formatted prompt
prompt = """Classify company {company} into one of the following sectors. Answer only with the
sector name: Technology, Consumer Cyclical, Industrials, Utilities, Healthcare, Communication,
Energy, Consumer Defensive, Real Estate, or Financial.
"""

# Loop through each company and classify it using OpenAI
for company in nasdaq100_ca["symbol"]:
    response = client.chat.completions.create(
        model="gpt-4o-mini",
        messages=[{"role": "user", "content": prompt.format(company=company)}],
        temperature=0.5
    )

    # Extract the sector from the response
    sector = response.choices[0].message.content

    # Assign the sector to the correct row in the DataFrame
    nasdaq100_ca.loc[nasdaq100_ca["symbol"] == company, "sector"] = sector

# Check the count of each sector
print(nasdaq100_ca["sector"].value_counts())

```

```

Technology      23
Consumer Cyclical  7
Healthcare      6
Energy          1
Consumer Defensive  1
Name: sector, dtype: int64

```

```
# Create a formatted prompt with the nasdaq100_ca DataFrame as a variable
prompt = f"""Provide summary information about Nasdaq-100 stock performance year to date (YTD)
of companies headquartered in CA, recommending the two best sectors and two or more companies
per sector.
Company data: {nasdaq100_ca}
"""

# Create an OpenAI request
response = client.chat.completions.create(
    model="gpt-4o-mini",
    messages=[{"role": "user", "content": prompt}],
    temperature=0.0
)

# Store and print the recommendations
stock_recommendations = response.choices[0].message.content
print(stock_recommendations)
```

Summary of Nasdaq-100 Stock Performance YTD for California-Based Companies

As of the current year, California-based companies in the Nasdaq-100 have shown varied performance across different sectors. The two best-performing sectors based on year-to-date (YTD) performance are **Technology** and **Consumer Cyclical**.

1. Technology Sector

The Technology sector has been the standout performer, with several companies achieving significant YTD gains. Here are two of the top companies in this sector:

- **Nvidia (NVDA)**
 - **YTD Performance:** 217.27%
 - Nvidia has been a leader in graphics processing units (GPUs) and artificial intelligence, driving substantial growth.
- **Meta Platforms (META)**
 - **YTD Performance:** 153.78%
 - Meta has seen a strong rebound as it continues to innovate in social media and virtual reality.

Other notable mentions in the Technology sector include:

- **Apple Inc. (AAPL)** - YTD: 43.00%
- **Adobe Inc. (ADBE)** - YTD: 57.23%
- **Advanced Micro Devices (AMD)** - YTD: 82.46%

2. Consumer Cyclical Sector

The Consumer Cyclical sector has also performed well, with several companies showing impressive YTD gains. Here are two top companies in this sector: